

Orienteering Problem

Optimizing Paths: **Balancing Time, Energy,
and Rewards**

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Outline

- 1 Introduction
- 2 Algorithm
- 3 Solution Approach
- 4 Variants of OP
- 5 Application of OP

Introduction

What is Orienteering Problem?

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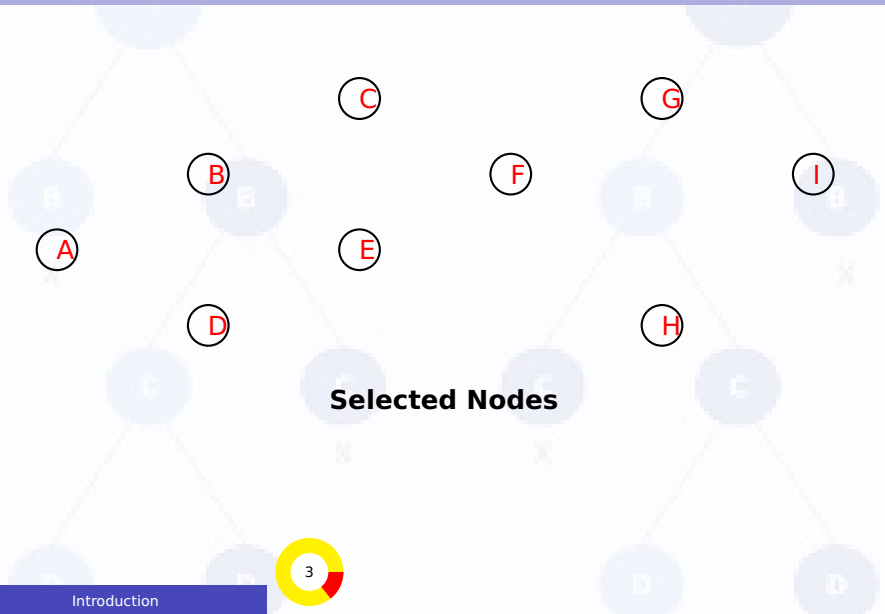
- Basically a routing problem

Introduction

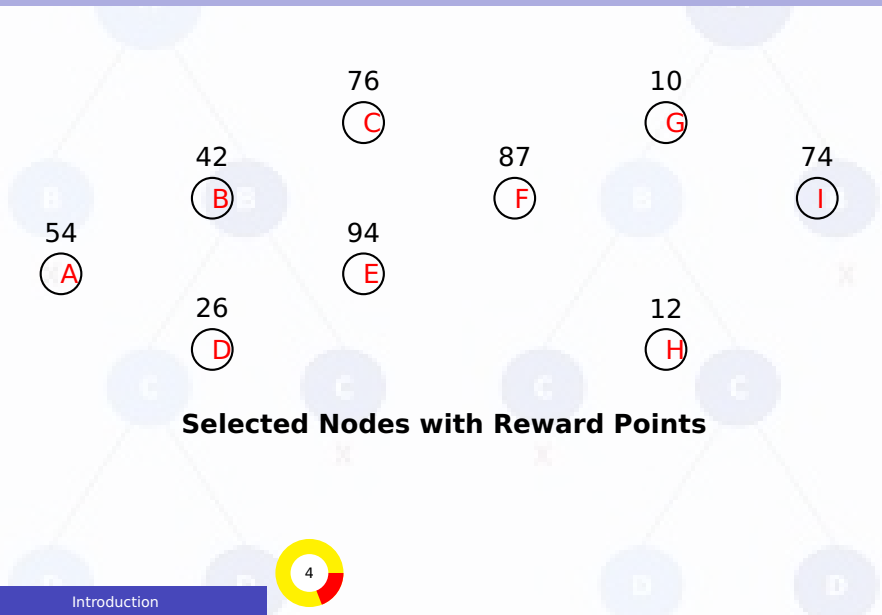
What is Orienteering Problem?

- Basically a routing problem
- Goal is to find the most rewarding path within the travel budget

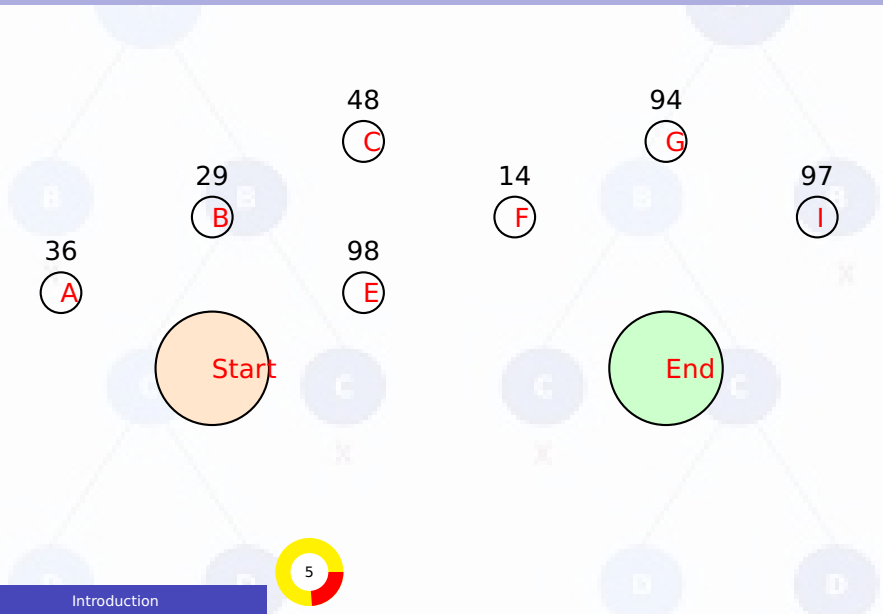
Understanding the problem..



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But there is a catch!!

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But there is a catch!!

- Limited Travelling Time

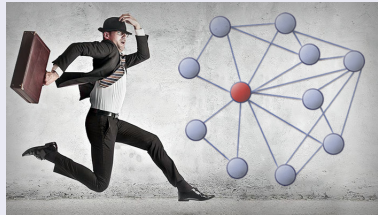
Diving into the Problem

Combination of two problems

Knapsack problem



Traveling Salesman



Key Points of Orientiering Problem

- Number of n locations with the given rewards

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- Number of n locations with the given rewards
- Initial location and final location

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- Number of n locations with the given rewards
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Heuristic algorithm

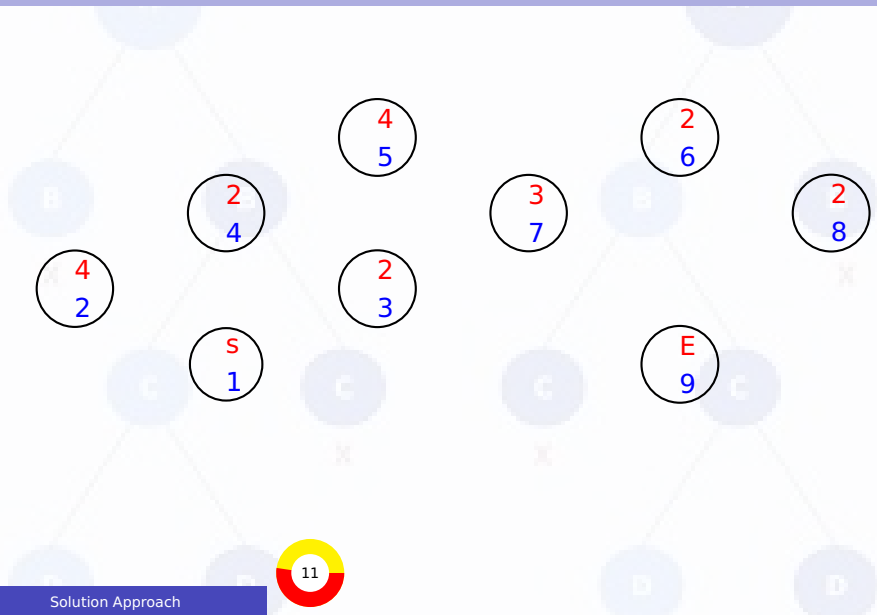
Two point exchange algorithm for the OP

```
For  $j$  = the first to the last point in  $path_{op}$ 
  For  $i$  = the first to the last point
    in the first to the last path in  $paths_{nop}$ 
    If exchanging  $i$  and  $j$  is feasible
      and the total score increases
      do the exchange and go to the A loop
    Else Set the
      best exchange = one with the highest score
  End B loop
  If the score of the
    best exchange  $\geq$  record - deviation
    make the best exchange
End A loop
```

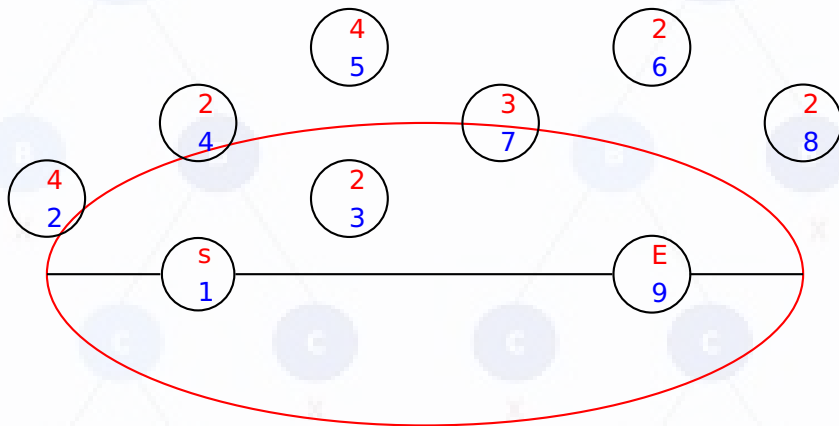
Equations for two points exchange

$$L(p) - (c_{jf_j} + c_{pj} - C_{pjf_j}) + \min_{\substack{k \text{ visited in } p, \\ k \neq 1, j}} (c_{i,k} + c_{i,pk} - C_{k,pk}),$$

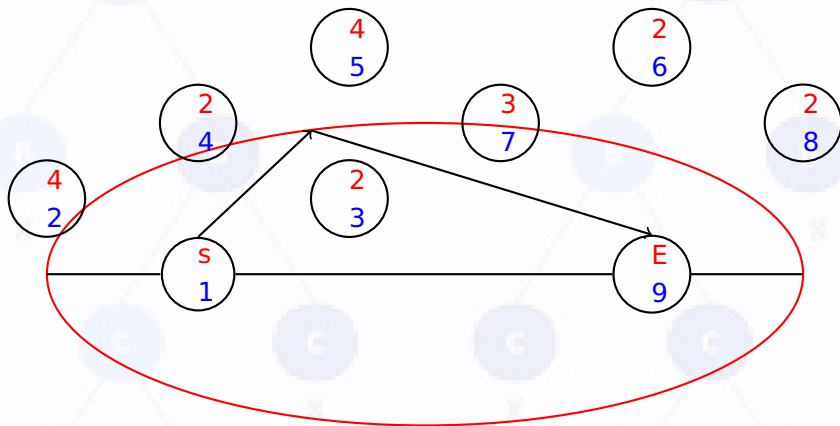
Heuristic algorithm



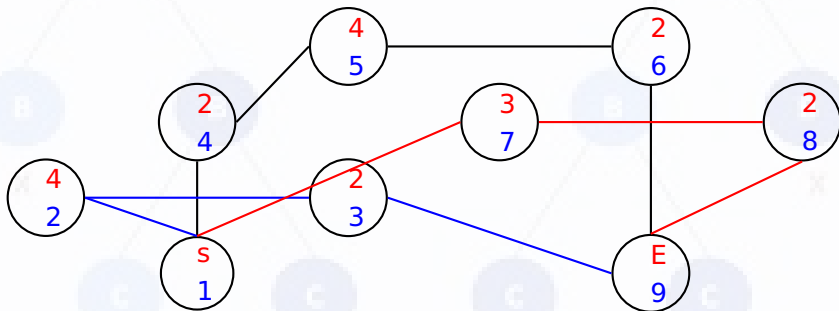
Heuristic algorithm



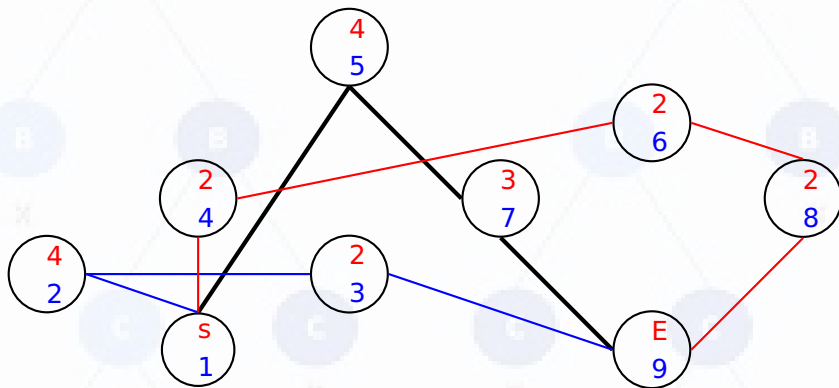
Heuristic algorithm



Heuristic algorithm



Heuristic algorithm



Heuristic Algorithm

One Point Exchange Algorithm

Algorithm for one-point movement in the OP

For i = the first to the last point in the T_{max} ellipse (say point i is in path q)

For j = the first to the last point in the first to the last path (p) in both $path_{op}$ and $paths_{nop}$ sets ($p \neq q$)

If inserting i in front of j on path p is feasible and the total score increases, then make the movement and go to the A loop

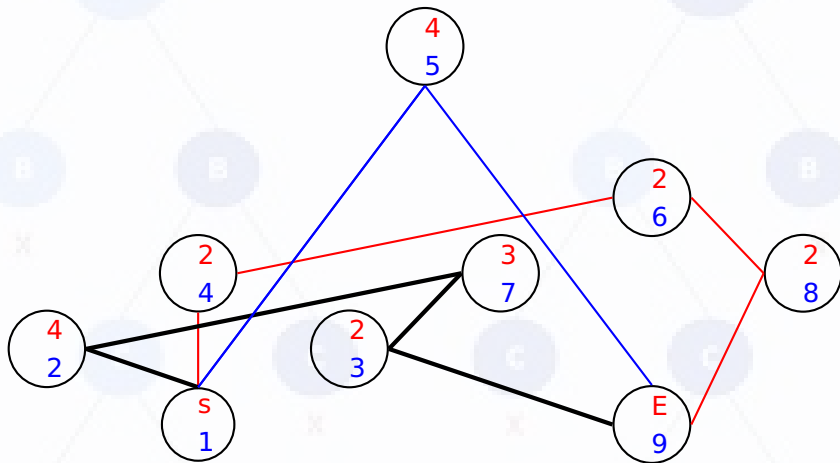
Else Set the best movement = one with the highest score

End B loop

If the score of the best movement $>$ record - deviation, then make the best movement

End A loop

Heuristic algorithm



Variants of OP

Team Orienteering Problem

Variants of OP

Team Orienteering Problem

- Extended version of the orienteering problem

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- Extended version of the orienteering problem
- Starting location contains many persons

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Time Depended OP Problem

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Time Depended OP Problem

- Time depended travel times

Variants of OP

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Time Depended OP Problem

- Time depended travel times
- Constraints

Efficient use of Drone



Figure: Applying OP on Drone

Military Navigation



Figure: Using OP for Navigation

Conclusion

*Thank you
for your patience*

For any further queries

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(Click for the link)